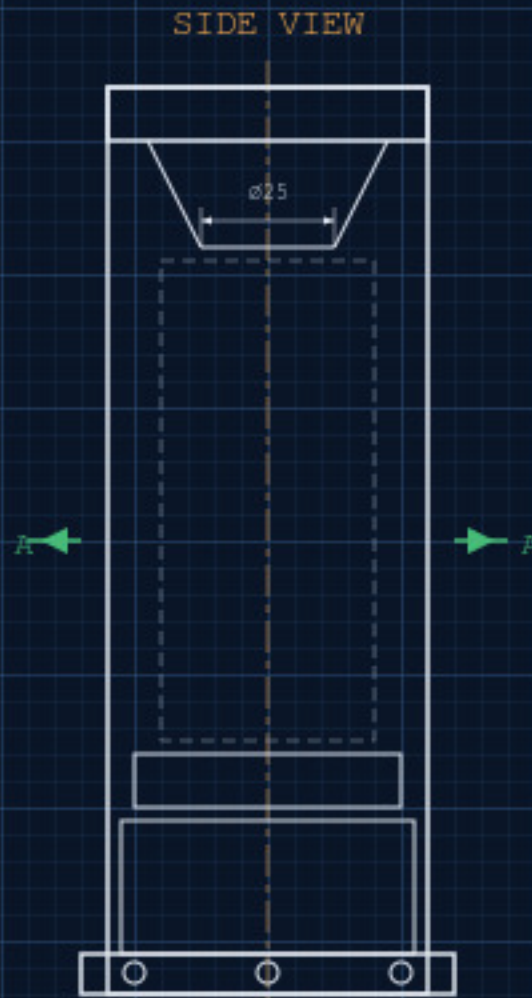
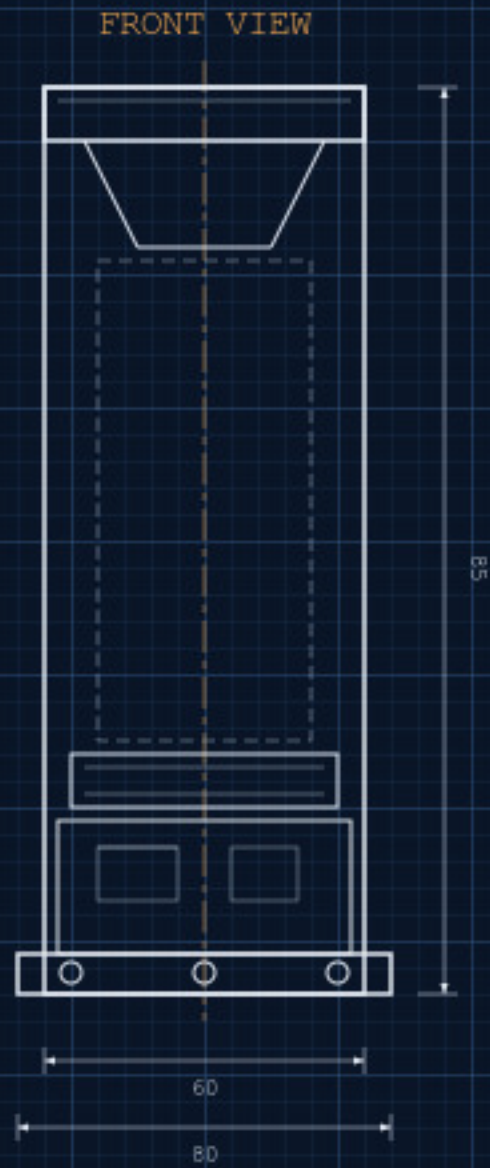
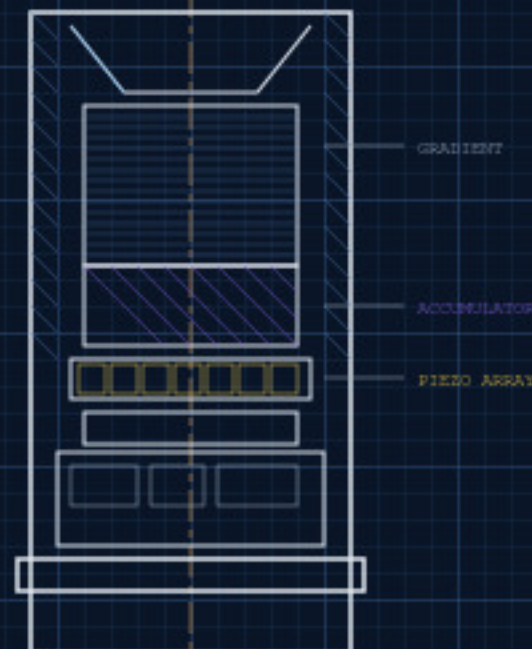




SECTION A-A



NOTES:

1. ALL DIMENSIONS IN MM
2. TOLERANCES: ± 0.1 UNLESS NOTED
3. MATERIAL: AL 6061-T6 HOUSING
4. SURFACE FINISH: $Ra\ 1.6\ \mu m$
5. BREAK ALL SHARP EDGES
6. REFER TO ATP-MK1-MTL-001 FOR CORE SPECIFICATION



AETHERION MARK 1

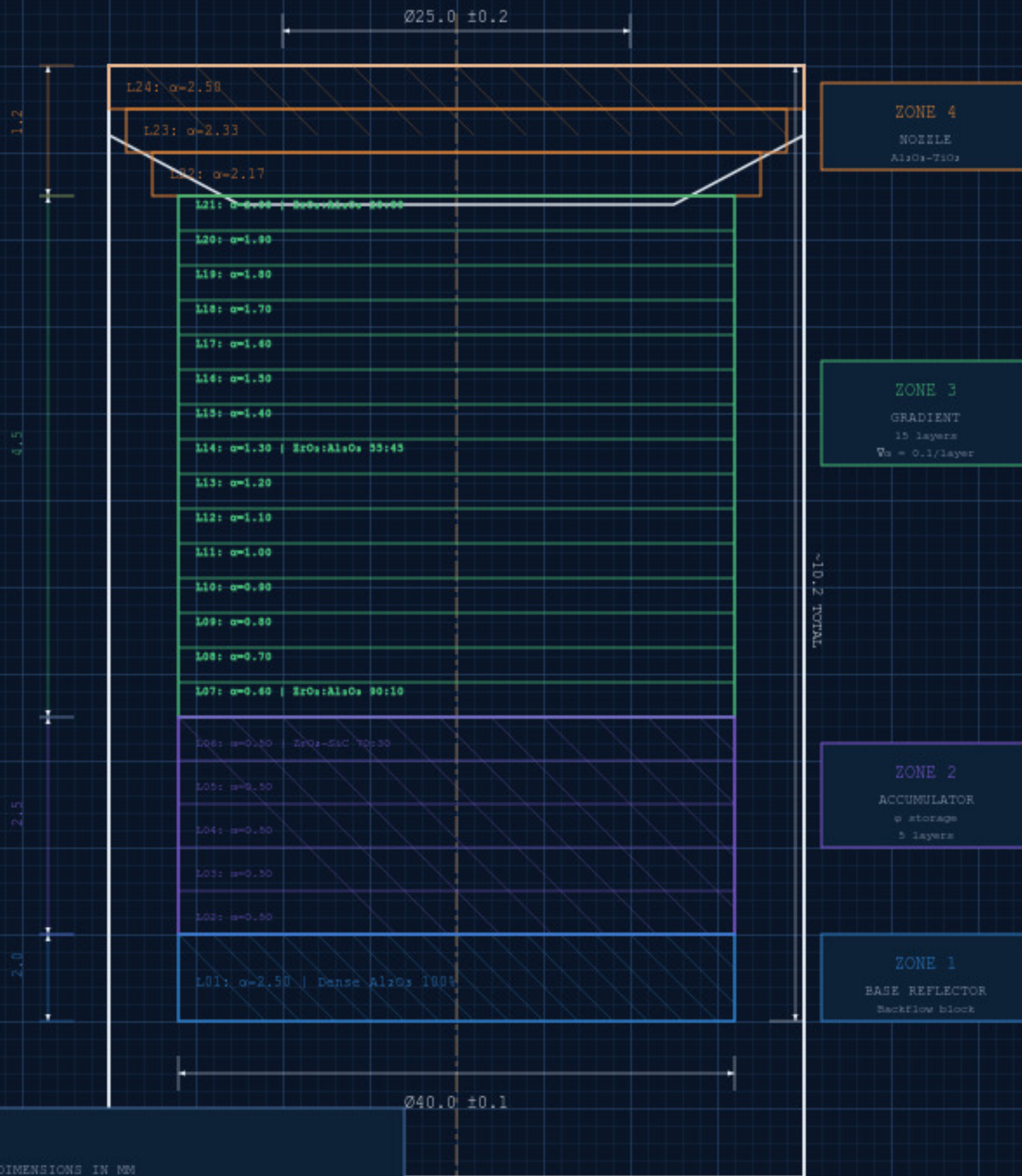
ORTHOGRAPHIC VIEWS

SCALE: 2:1

DWG: ATP-MK1-BP-001

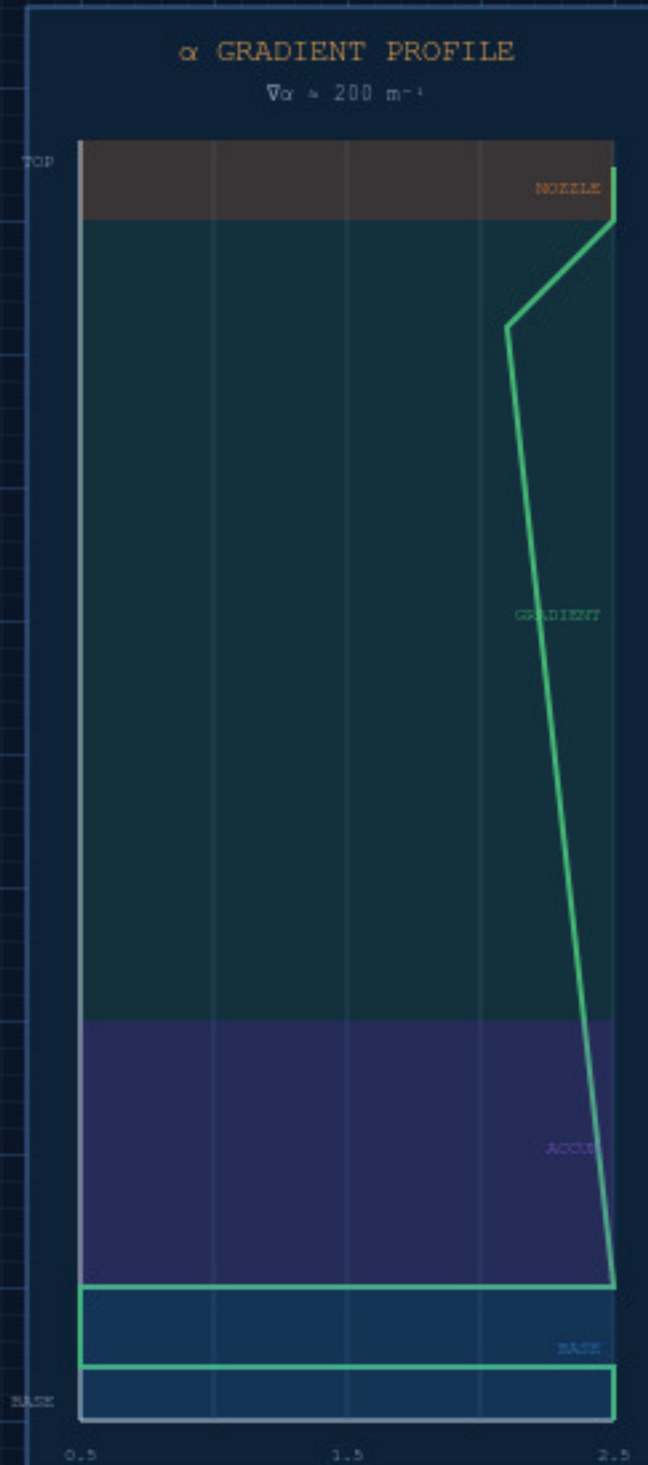
REV: 1.0

METAMATERIAL CORE - CROSS SECTION (SCALE 5:1)



NOTES:

- ALL DIMENSIONS IN MM
- α VERIFIED BY ϵ MEASUREMENT
- SEE ATP-MK1-MTL-001 FOR COMPLETE FABRICATION SPEC



MATERIAL SUMMARY

NOZZLE:	$\text{Al}_2\text{O}_3\text{-TiO}_2$ (graded 75:25 ~ 25:75)
GRADIENT:	$\text{ZrO}_2\text{-Al}_2\text{O}_3$ (graded 90:10 ~ 20:80)
ACCUM:	$\text{ZrO}_2\text{-SiC}$ (70:30 constant)
BASE:	Dense Al_2O_3 (100%)

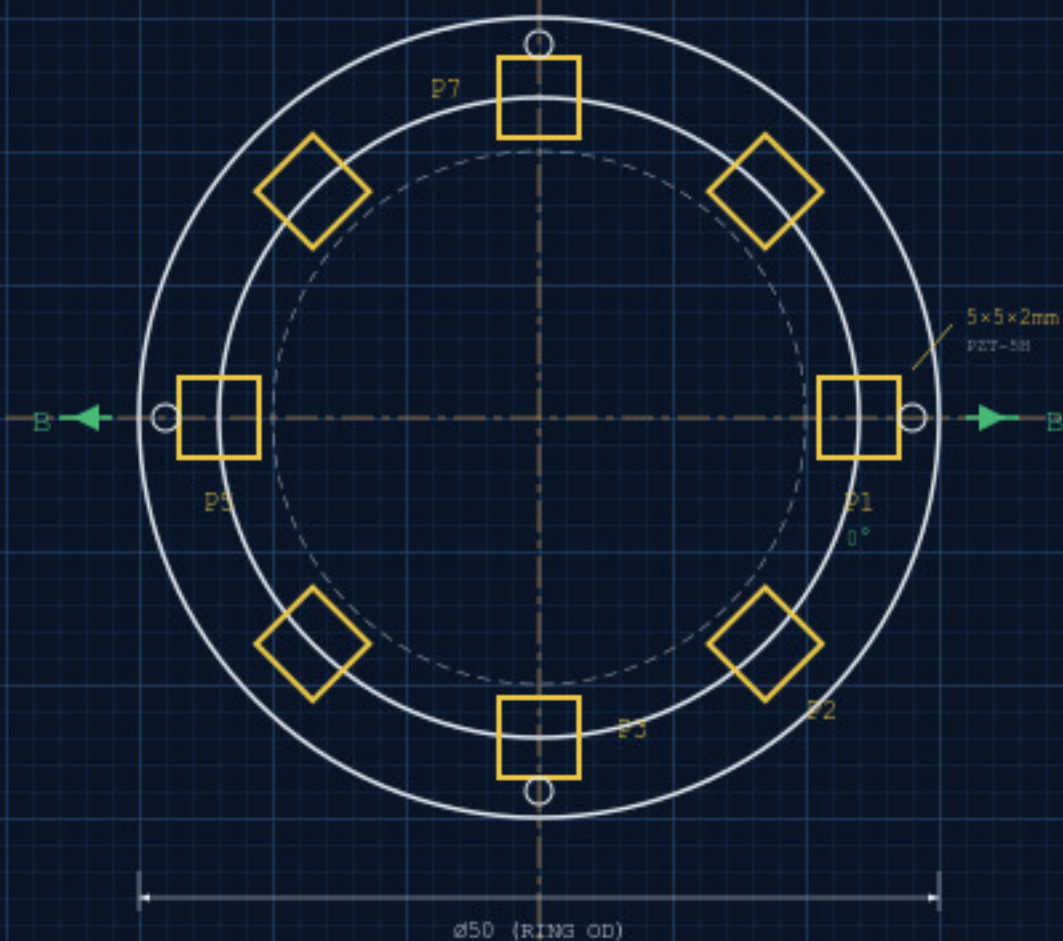
AETHERION MARK 1

METAMATERIAL CORE DETAIL DWG: ATP-MK1-BP-002

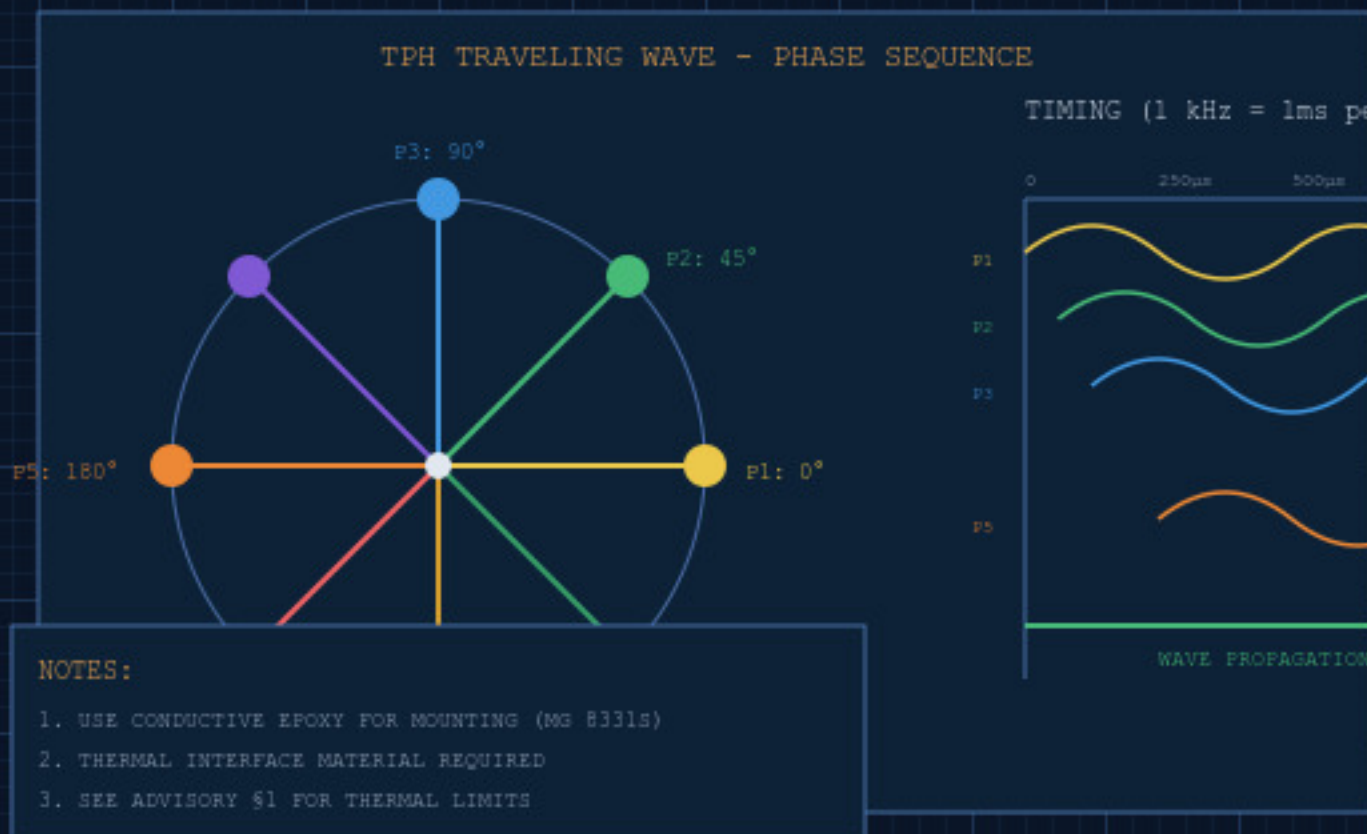
SCALE: 5:1

REV: 1.0

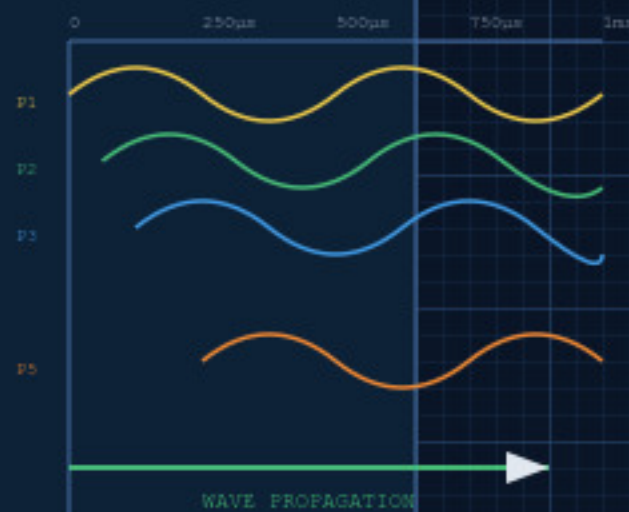
TOP VIEW - PIEZO RING



PHASE DIAGRAM (TPH MODE)



TIMING (1 kHz = 1ms period)



SECTION B-B



PIEZO SPECIFICATIONS

PARAMETER	VALUE
Material	PZT-5H (Navy Type VI)
Dimensions	5 x 5 x 2 mm
Quantity	8 (radial array)
d_{33} Coefficient	>600 pC/N
Max Voltage	200V (100V/mm)
Curie Temp (T_c)	195°C (CRITICAL)
Operating Limit	90°C MAX
Phase Spacing	45° (360°/8)

AETHERION MARK 1

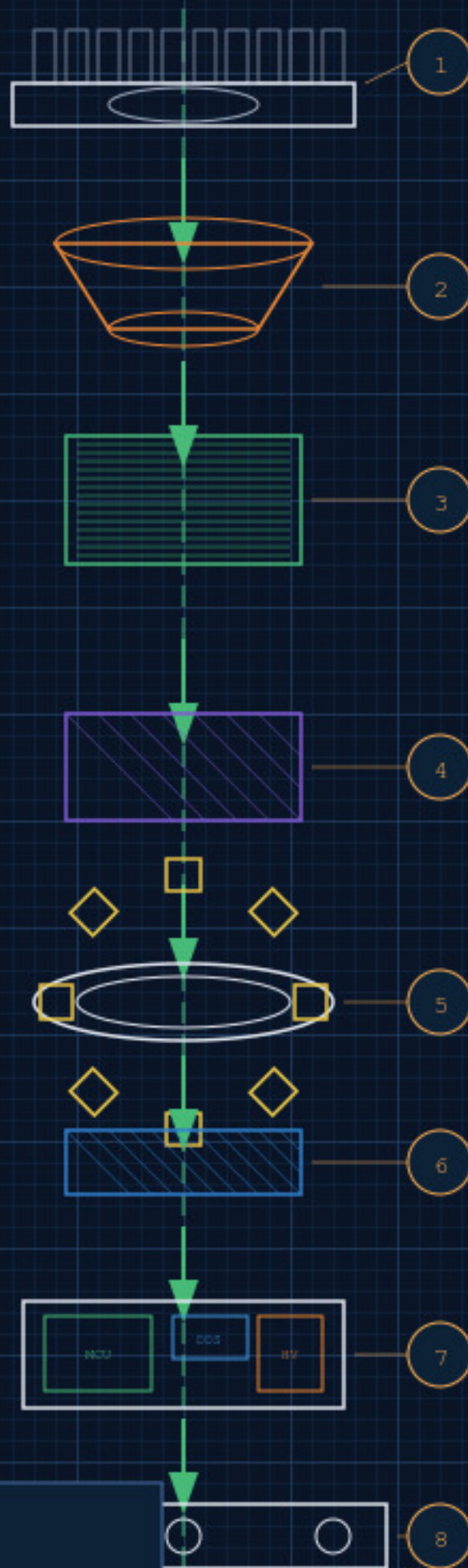
PIEZO ARRAY ASSEMBLY

DWG: ATP-MK1-BP-003

SCALE: 3:1

REV: 1.0

EXPLODED ASSEMBLY - ALL COMPONENTS



PARTS LIST

ITEM	PART NAME	MATERIAL	QTY
1	TOP COVER (w/ heatsink fins)	Al 6061-T6	1
2	NOZZLE CONE (3 layers, $\alpha=2.0-2.5$)	$Al_2O_3-TiO_2$	1
3	GRADIENT STACK (15 layers, $\alpha=0.6-2.0$)	$ZrO_2-Al_2O_3$	1
4	ACCUMULATOR (5 layers, $\alpha=0.5$)	ZrO_2-SiC	1
5	PIEZO RING ASSY (8x actuators on ring)	PET-5H	1
6	BASE REFLECTOR (1 layer, $\alpha=2.5$)	Dense Al_2O_3	1
7	ELECTRONICS BAY (MCU, DDS, HV amps)	PCB + ICs	1
8	MOUNTING FLANGE (3x M4 mounting holes)	Al 6061-T6	1

FASTENERS: M2x8 SHCS (24), M3x12 SHCS (16), M4x16 (3)

ASSEMBLY SEQUENCE

1. Install MOUNTING FLANGE (8)
2. Install ELECTRONICS BAY (7)
3. Install BASE REFLECTOR (6)
4. Install PIEZO RING ASSY (5)
5. Stack ACCUMULATOR (4) + GRADIENT (3)
6. Install NOZZLE CONE (2)
7. Install TOP COVER (1) with TIM

NOTES:

1. APPLY THERMAL INTERFACE MATERIAL BETWEEN (1) AND (5)
2. CORE ASSEMBLY (2-4-3-6) IS PRE-BONDED UNIT
3. VERIFY PIEZO WIRING BEFORE FINAL CLOSURE

AETHERION MARK 1

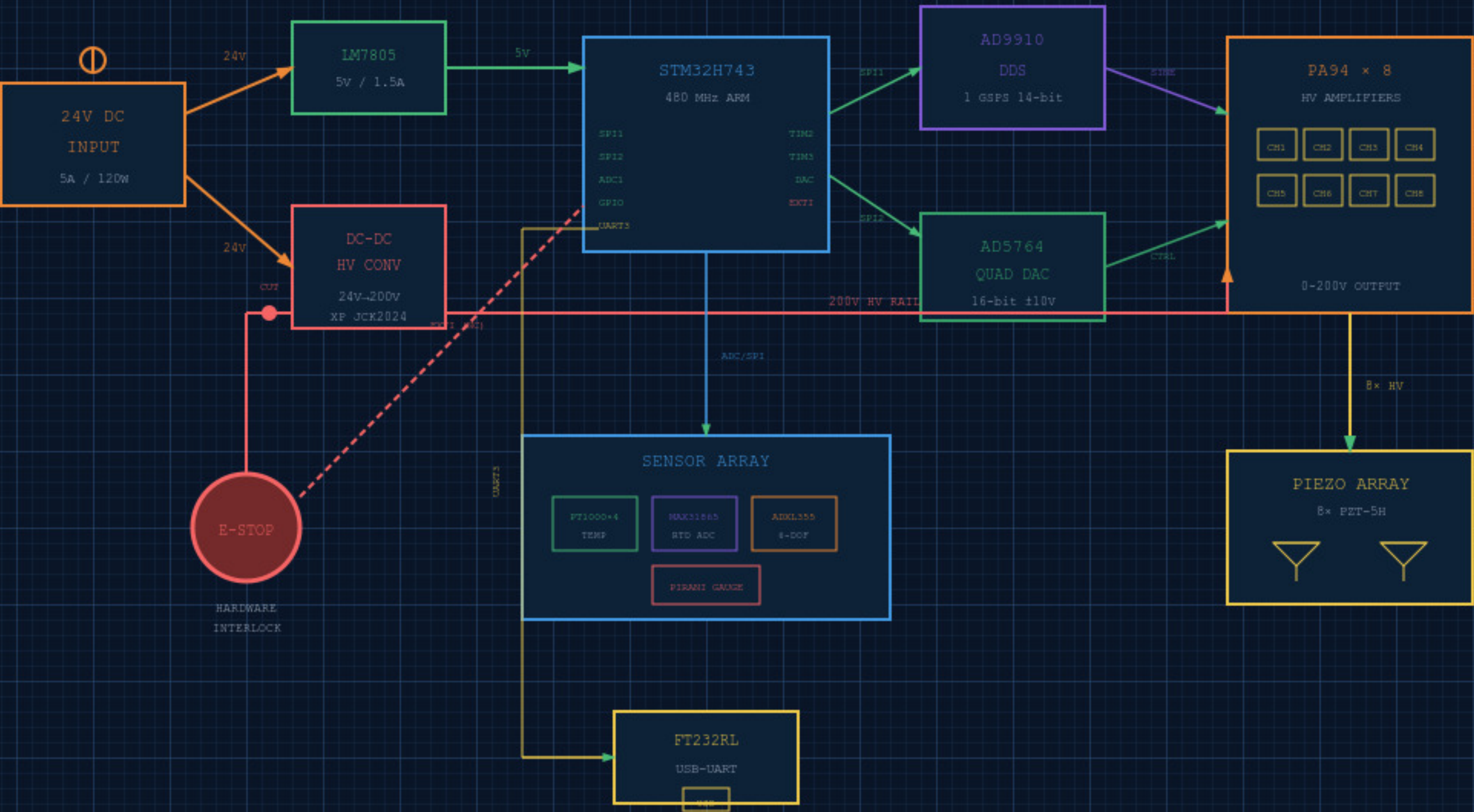
EXPLODED ASSEMBLY VIEW

DWG: ATP-MK1-BP-004

SCALE: NTS

REV: 1.0

ELECTRONICS SYSTEM BLOCK DIAGRAM



LEGEND

- POWER (24V/200V)
- SIGNAL (5V/LOGIC)
- SAFETY INTERLOCK

POWER BUDGET

MCU + Logic:	~2W	5V Rail:	1.5A max
DDS + DAC:	~3W	200V Rail:	250mA max
HV Amps (peak):	~40W	24V Input:	5A max
TOTAL (peak):	~50W		

AETHERION MARK 1

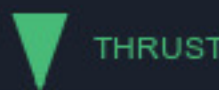
ELECTRONICS BLOCK DIAGRAM DWG: ATE-MK1-BP-005

SCALE: NTS

REV: 1.0

AETHERION MARK 1

Cross-Section Technical Drawing



SPECIFICATIONS

Total Mass: 250g
Core Mass: 50g
Layers: 23
 $\nabla\alpha$: 200 m^{-1}
Thrust: 100-500 nN
Freq: 1-10 kHz
Power: 5-50 W
HV: 200V DC

LEGEND

- Nozzle ($\alpha=2.5$)
- Gradient ($\alpha:2\rightarrow0.5$)
- Core ($\alpha=0.5$)
- Piezo Array
- Reflector ($\alpha=2.5$)
- Electronics

85 mm

TOP COVER (Al)

NOZZLE

$\alpha = 2.5$

Aperture $\varnothing 25\text{mm}$

GRADIENT ZONE

$\alpha: 2.0 \rightarrow 0.5$

(15 metamaterial layers)

Energy Flow $\downarrow$

Core: 30mm

ACCUMULATOR CORE

$\alpha = 0.5 \mid \varphi_{\text{max}}$

Zero-Point Energy Storage

$\varphi_{\text{max}} = 0.108$

BASE REFLECTOR ($\alpha = 2.5$)

ELECTRONICS BAY

MCU

DDS

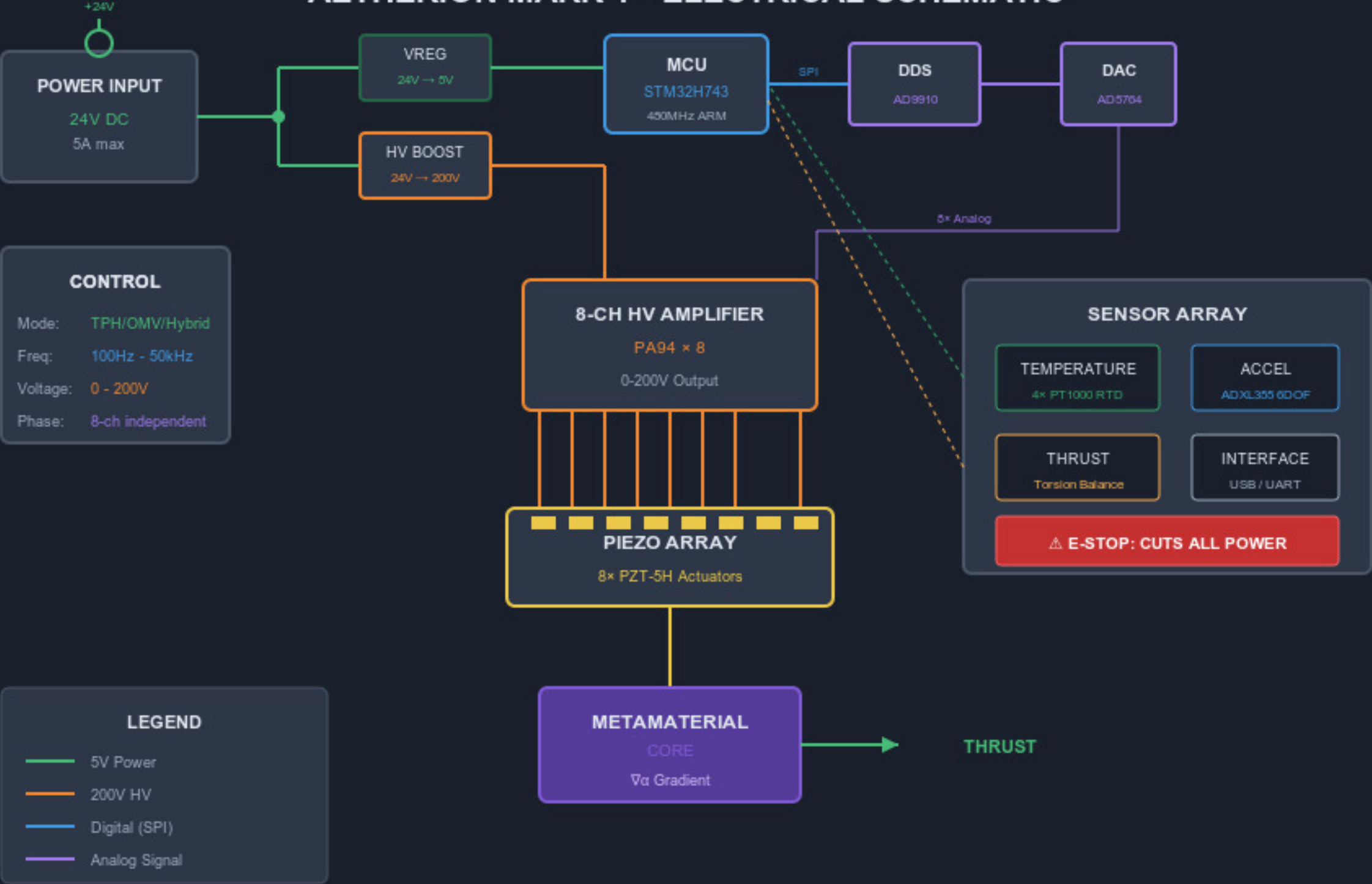
HV AMP

PSU

MOUNTING FLANGE

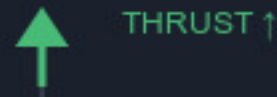
60 mm

AETHERION MARK 1 - ELECTRICAL SCHEMATIC



AETHERION MARK 1

Exploded Assembly View



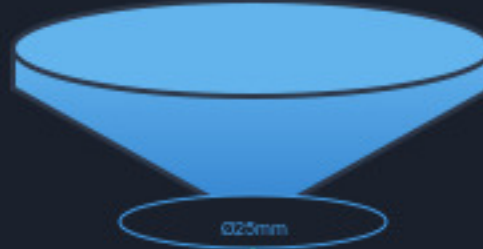
ASSEMBLY ORDER

- 8 → Flange first
- 7 → Electronics
- 6 → Base
- 5 → Piezo ring
- 4 → Accumulator
- 3 → Gradient stack
- 2 → Nozzle last



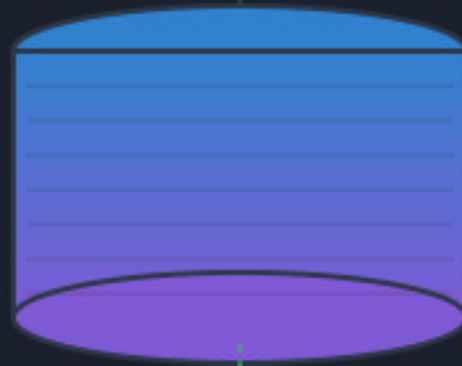
1. TOP COVER

Aluminum 6061-T6



2. NOZZLE CONE

Metamaterial $\alpha=2.0 \rightarrow 2.5$



3. GRADIENT STACK

15 layers, $\alpha: 2.0 \rightarrow 0.5$

$\nabla\alpha = 200 \text{ m}^{-1}$



4. ACCUMULATOR

$\alpha = 0.5$ (constant)

ZPE Storage Zone



5. PIEZO ARRAY

8× PZT-5H actuators

Phase-controlled, 200V



6. BASE REFLECTOR

$\alpha = 2.5$ (prevents backflow)



7. ELECTRONICS BAY

STM32H7, AD9910, PA94×8

TPH / OMV PROPULSION SYSTEM

Temporal Pulse Hierarchy + Oscillatory Modulation of Vacuum

TPH MODE

Pulsed Asymmetric Thrust



STATIC $\nabla\alpha$



Symmetric (No thrust)

TRAVELING ∇L



Piezo shock wave $\rightarrow$

OMV MODE

Continuous Ponderomotive Thrust



Ponderomotive Rectification

$$F \propto (\nabla a)^2 = (\nabla a_0)^2 \cdot \cos^2(\omega t)$$

$$\cos^2(\omega t) = \frac{1}{2}[1 + \cos(2\omega t)]$$

↑ DC Component = Net Thrust

OPERATING MODE COMPARISON

Parameter	TPH Mode	OMV Mode	Hybrid
Waveform	Sharp Pulse	Sine Wave	Modulated Pulse
Thrust Type	Impulse (pN·s)	DC Force (pN)	Both
Magnitude	123 pN·s/pulse	197 pN DC	~500 nN @ 10kHz
Application	Maneuvering	Station-keeping	Full flight

PIEZO RING (Top View)



Firing Sequence:

$t=0$: P1,P2 $\rightarrow$

$t=1/4$: P3 $\rightarrow$

$t=1/2$: P4,P5 $\rightarrow$

Traveling wave CW

THRUST vs FREQUENCY



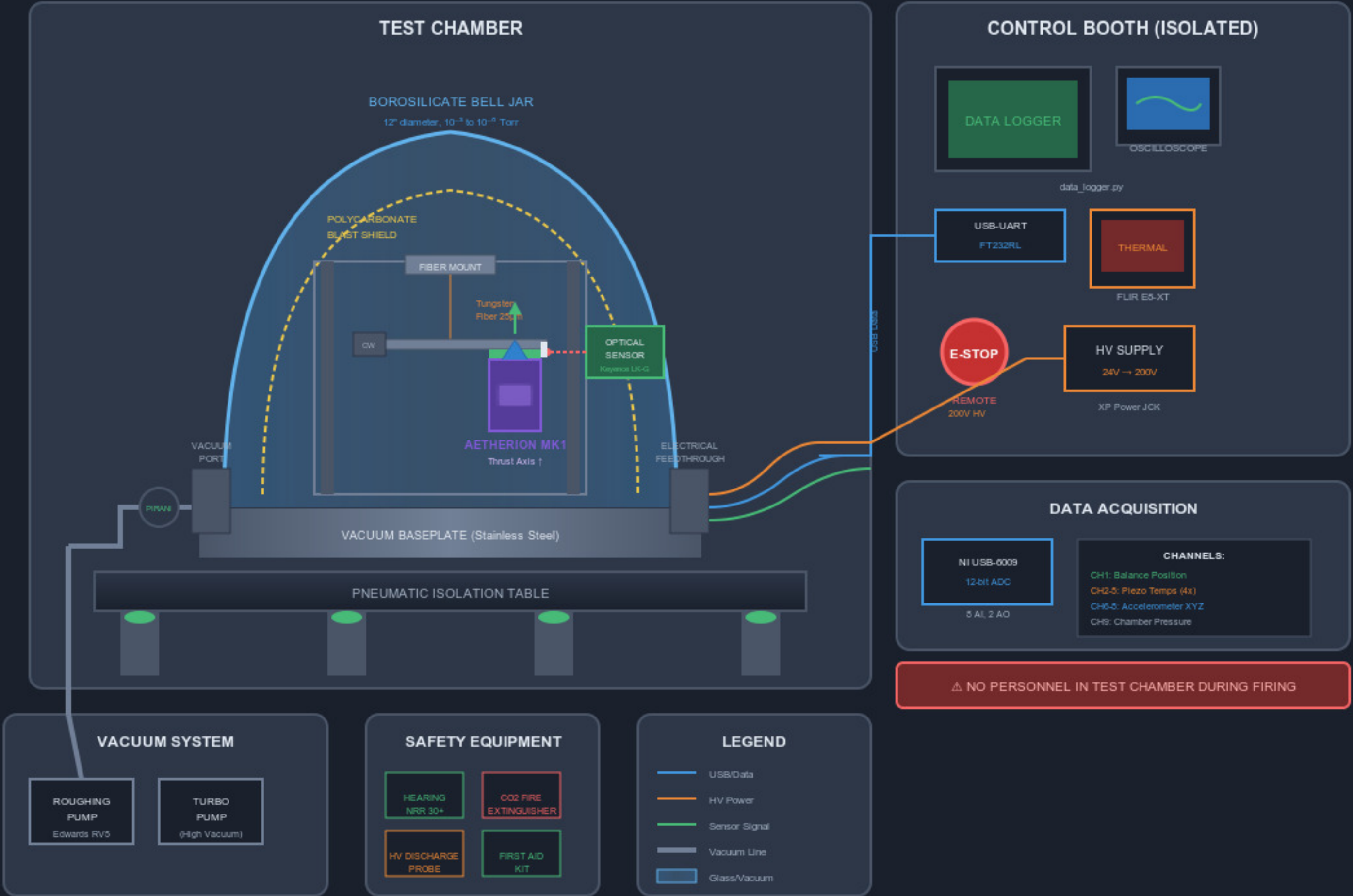
Results:

$$F = 50 \text{ nN/kHz}$$

Monte Carlo: ✓

AETHERION MARK 1 - TEST SETUP

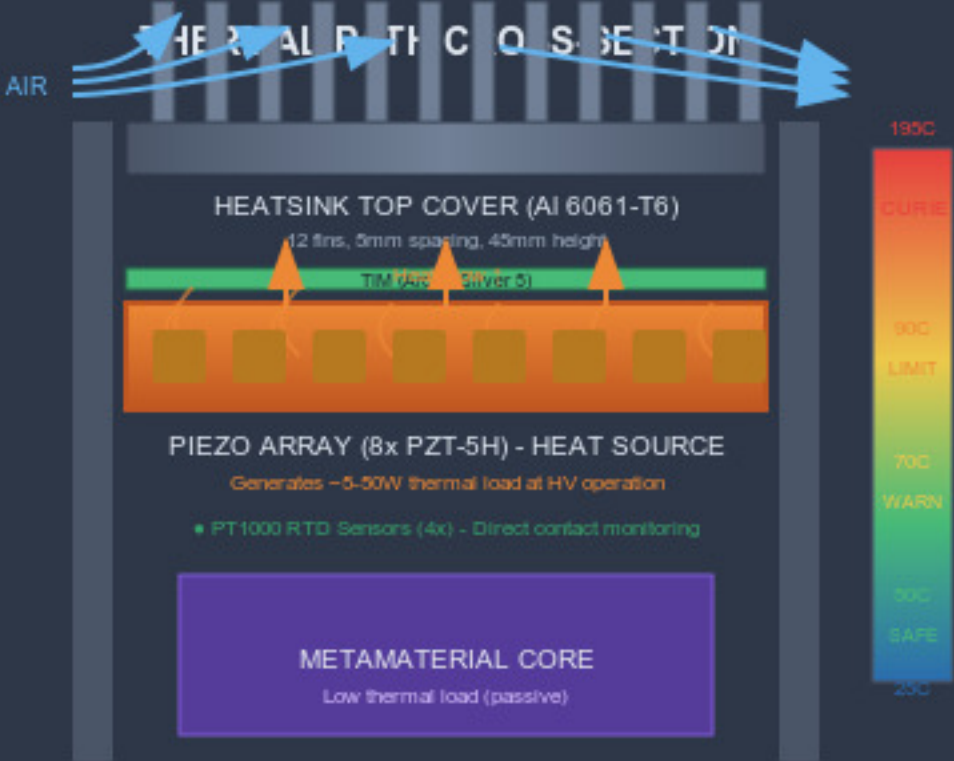
Torsion Balance + Vacuum Chamber Configuration



THERMAL MANAGEMENT SYSTEM

Red Team Advisory Compliance - Thermal Depolarization Prevention

⚠ CRITICAL: PZT-5H Curie Temperature = 195C
Operating Limit: 90C | Safe: below 70C | Depolarization = Permanent Failure



THERMAL BUDGET

Source	Power	Mitigation
Piezo Dielectric Loss	~30W	Heatsink fins
Piezo Mechanical Loss	~15W	TIM contact
HV Amplifier (PA94)	~5W	Separate heatsinks
TOTAL	~50W	Duty cycle limit

DUTY CYCLE THERMAL CONTROL



Duty Cycle = 10s / 70s = 14.3% MAX

MCU enforces this automatically (hardcoded)

Manual override PROHIBITED

TEMPERATURE RESPONSE



THERMAL MANAGEMENT COMPONENTS

Component	Specification	Purpose	BOM ID
Heatsink Top Cover	Al 6061-T6, 12 fins, 45mm	Primary heat dissipation	MC-001
Thermal Paste	Arctic Silver 5, 8.9 W/mK	Piezo-heatsink interface	MC-005
PT1000 Sensors (4x)	Class A, 2mm probe	Real-time temp monitoring	SN-001